



INSTITUTO GALEGO
DE FÍSICA
DE ALTAS ENERXÍAS



Current work and plans for the inclined air showers reconstruction

on behalf of the Santiago HAS Team

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Introduction

- HAS reconstruction relies on fitting a 2D muon map ρ_μ to the SD footprint:

$$\rho_\mu = N_{19}(A, E, \theta) \rho_\mu^{\text{ref}}(x, y, \theta, \varphi)$$

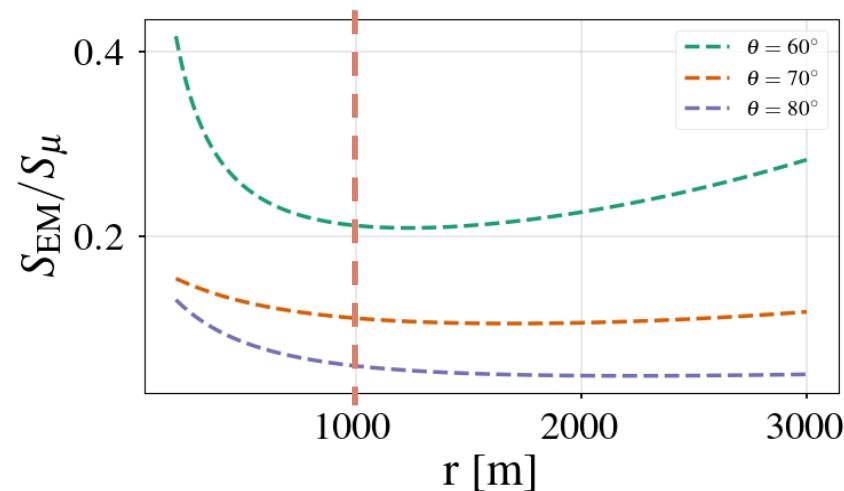
where:

- $\rho_\mu^{\text{ref}}(x, y, \theta, \varphi)$: reference map
 - expected n_μ/m^2 at (x, y) for a 10^{19} eV proton shower with direction (θ, φ) (from QGSJetII-03)
 - parameterized in θ and φ ; **shape fixed during the fit**
 - N_{19} : scaling factor (only free parameter)
 - serves as a proxy for the shower size
- Assumption: $\rho_\mu^{\text{ref}}(x, y, \theta, \varphi)$ is universal, with only weak dependence on energy, hadronic model and primary mass

Electromagnetic correction

$$S_{\mu}^{\text{meas}} = \frac{S^{\text{meas}}}{1 + R_{\text{EM}}/\mu}$$

$$\frac{S_{\text{EM}}}{S_{\mu}} = \frac{S_{\text{EM}}^{\text{pure}}}{f_{\mu\text{res}} S_{\mu}} + \frac{S_{\text{EM}\mu}}{S_{\mu}} + \frac{S_{\text{EM}\pi}}{S_{\mu}}$$



To do:

- Explore the signal ratio based in Universality
- Compare with legacy (older) correction
- Compare with simulated signals for different models
- Explore EM correction from SSD ?
- Re-parameterize with new hadronic models (EPOS-LHC R, newest Sybill versions)?

Current HAS work → Observer production (icrc2025-test7) [cutting edge]

Simulations: SD1500 → Standard application: **SdHASSimulationReconstruction**

Showers: Corsika-77410, **EPOS-LHC**, **Proton**, $\log(E / \text{eV}) = (19, 19.6)$, $\theta = (50^\circ, 89^\circ)$, triggers T5_HAS & 6T5

→ Angular and axis reconstruction, goodness of the fit, shower size [Marina]

→ Study impact of changing xml configuration for reconstruction → Two shower front models (*Sphere and MuonArrival*) and both EM corrections [Francisco]

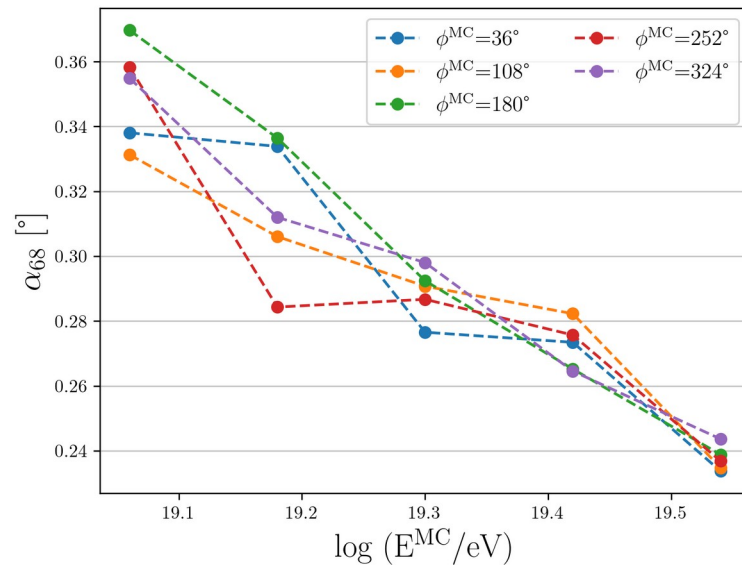
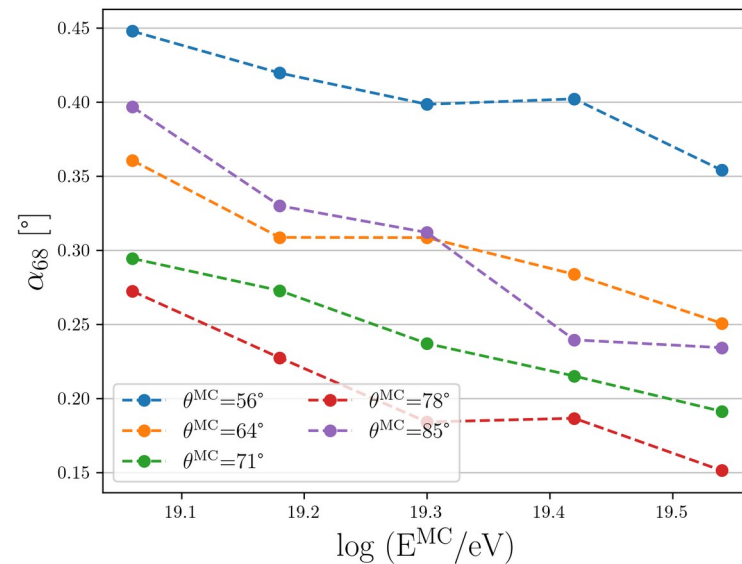
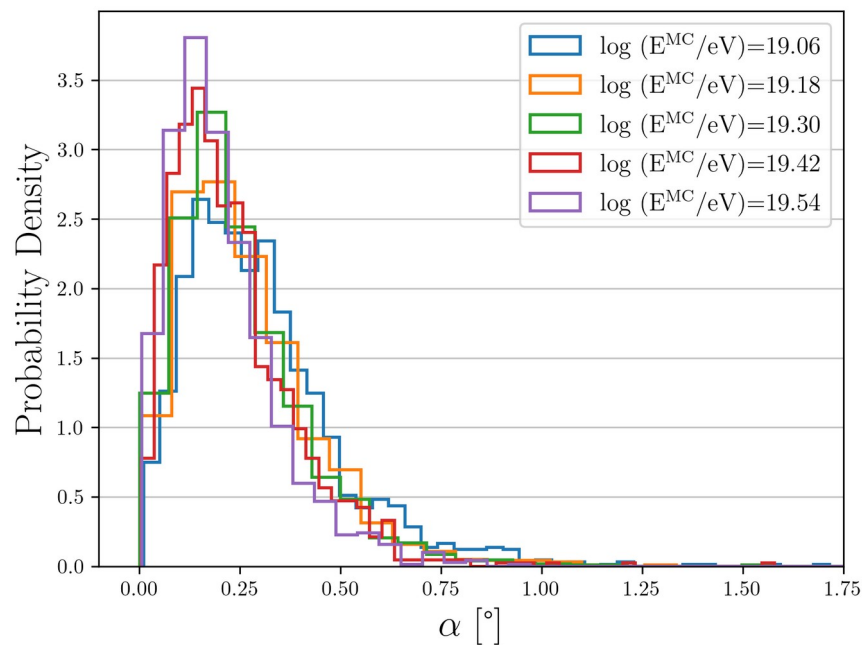
Data: standard ADST cuts `<offline_src>/ADST/Analysis/cuts/ICRC2025/sd{1500, 750}_inclined.cfg`

→ Average 2D signal map in bins of (θ, ϕ) for SD1500 and SD750 Phase I (goal: compare with simulations) [Joaquin]

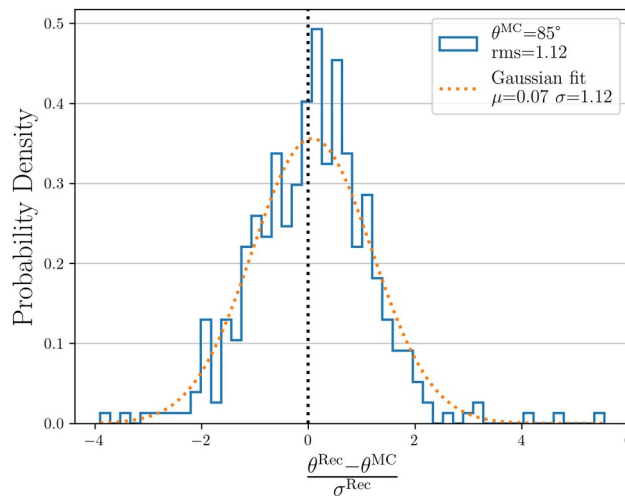
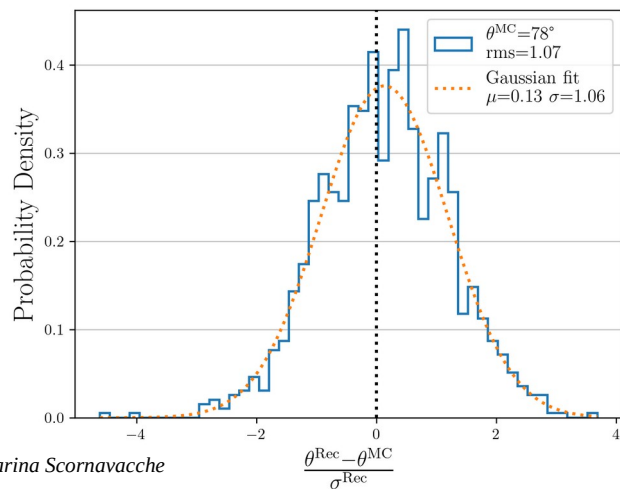
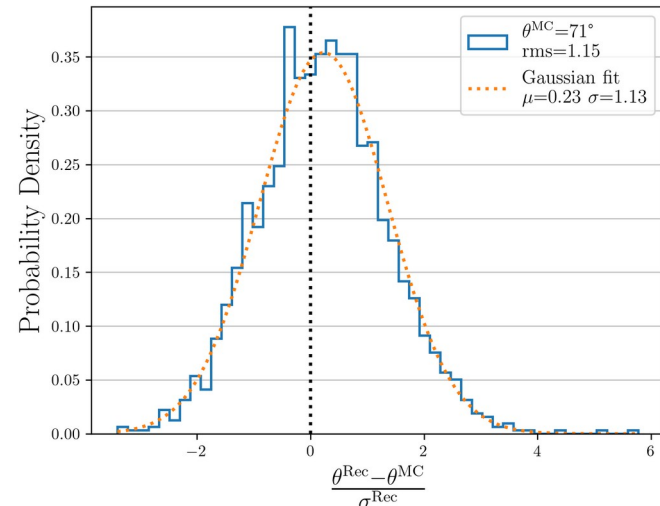
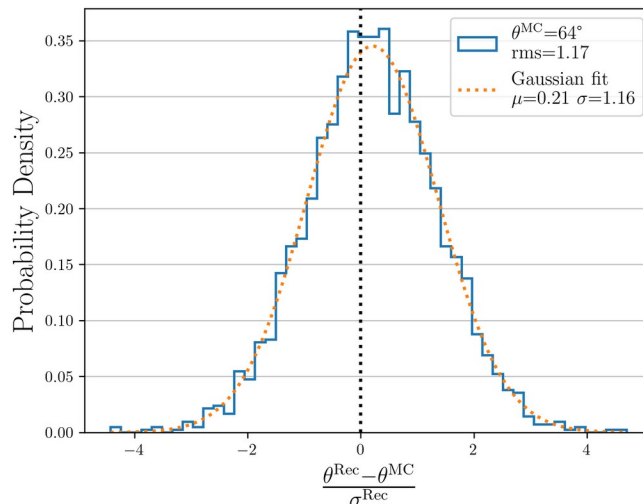
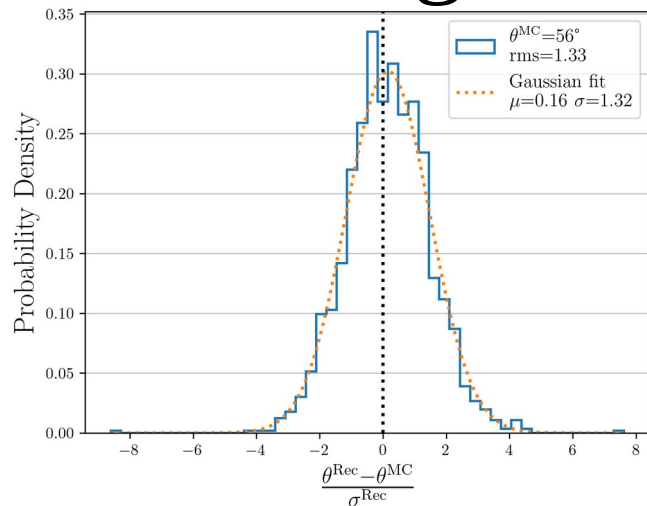
→ Validate Phase I and Phase II HAS event-by-event [Fernando]

Axis reconstruction

$$\cos \alpha = \frac{\vec{r}_{\text{Rec}} \cdot \vec{r}_{\text{MC}}}{\|\vec{r}_{\text{Rec}}\| \|\vec{r}_{\text{MC}}\|}$$

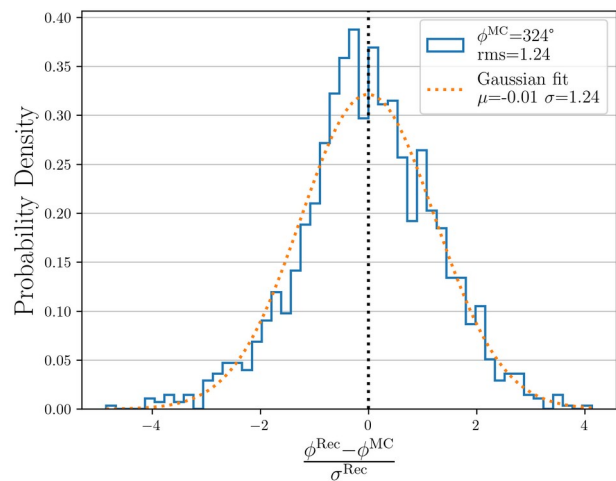
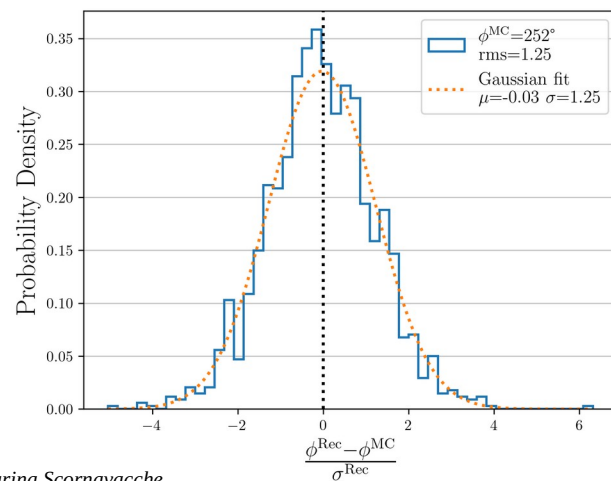
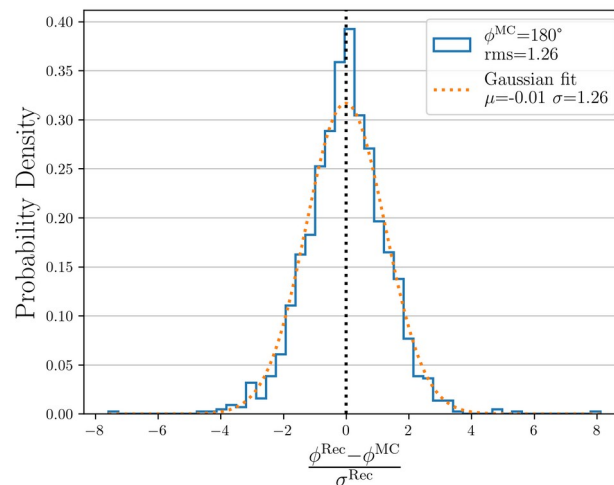
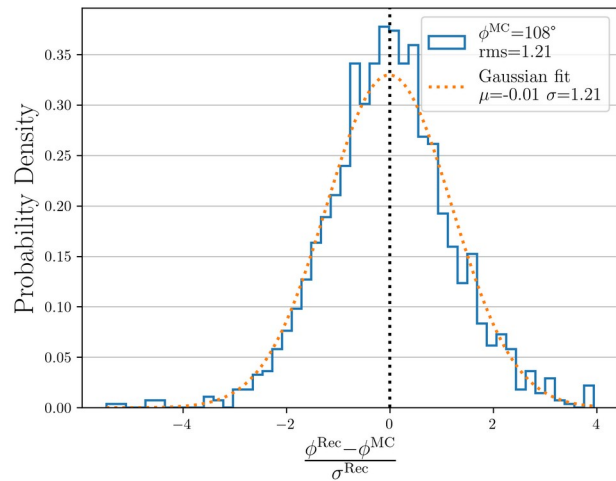
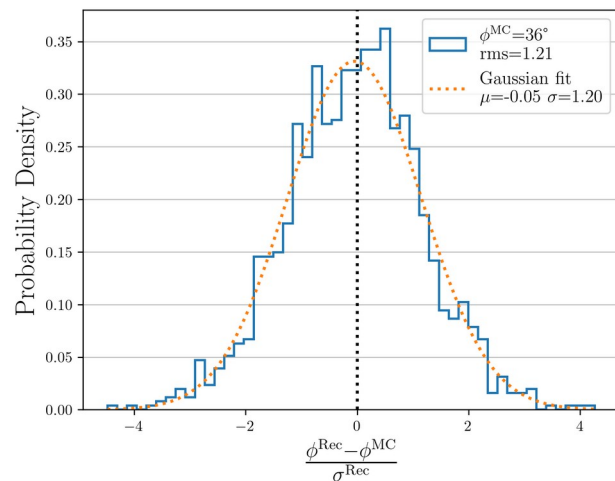


Zenith angle reconstruction



- slight bias
- underestimated uncertainties for $\theta = 56^\circ$
- no strong tails

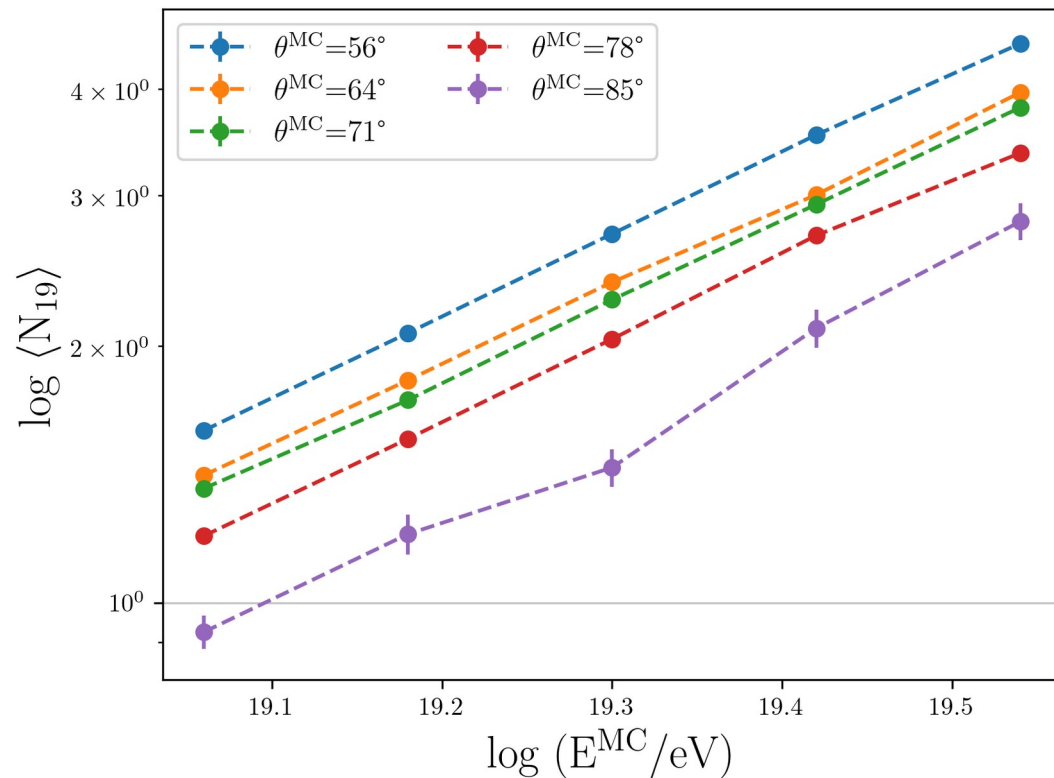
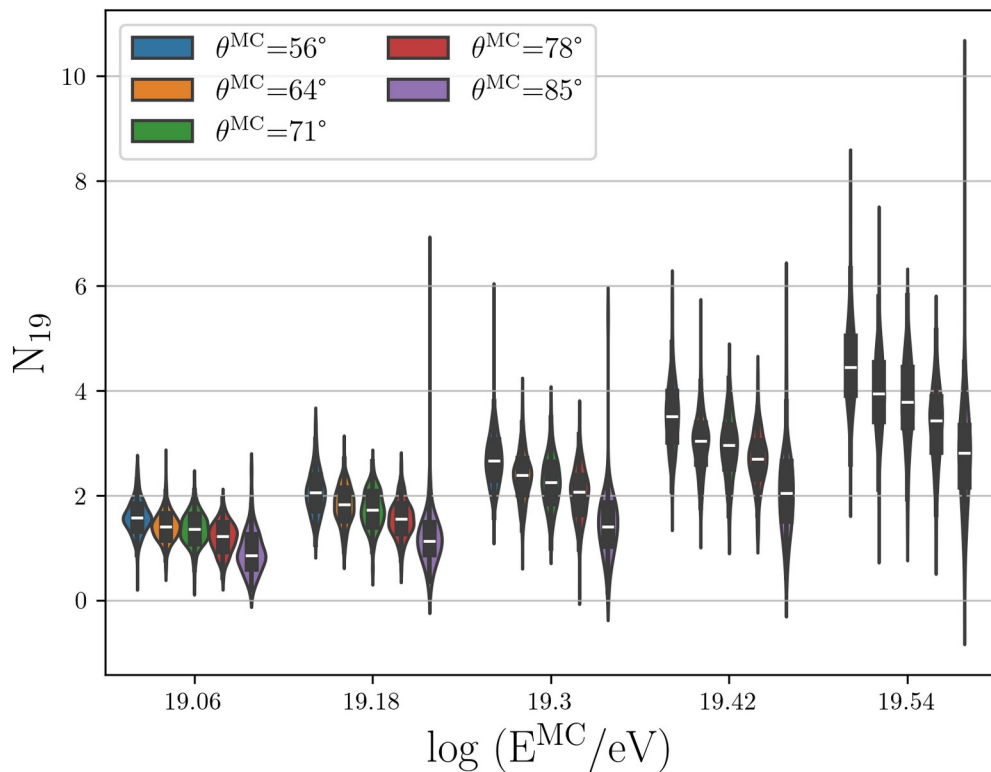
Azimuth angle reconstruction



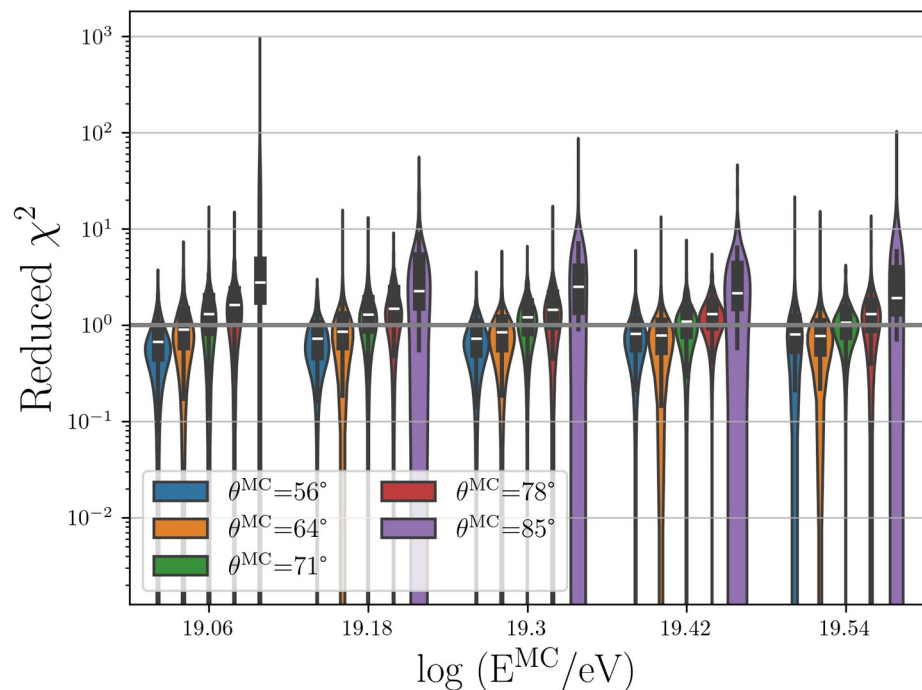
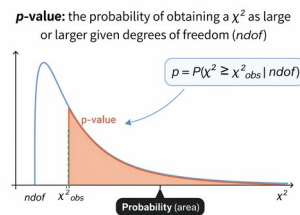
- no bias or strong tails
- underestimated uncertainties

Shower size

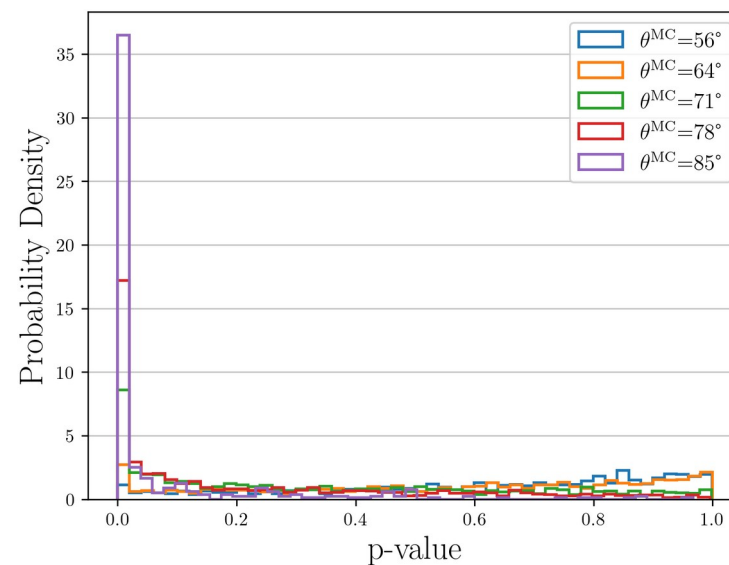
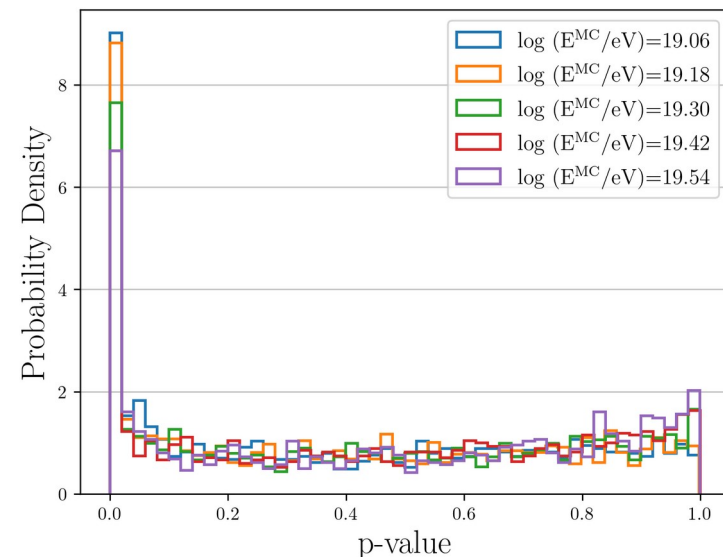
$$\langle N_{19} \rangle_{1\text{st bin}} \sim 1.37 \text{ [1.22 paper]}$$



Goodness of the fit



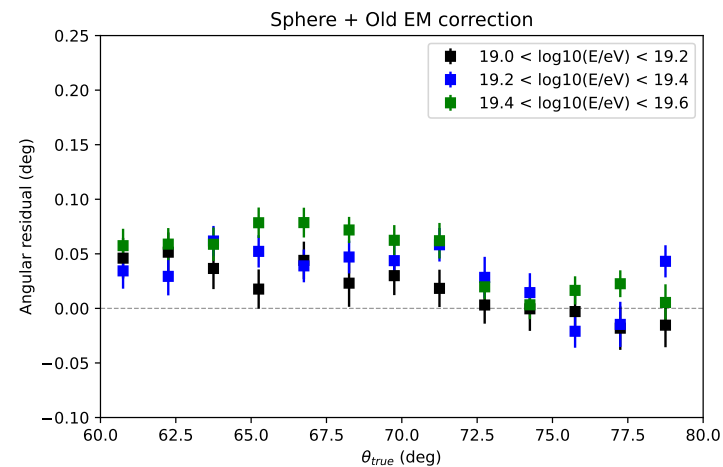
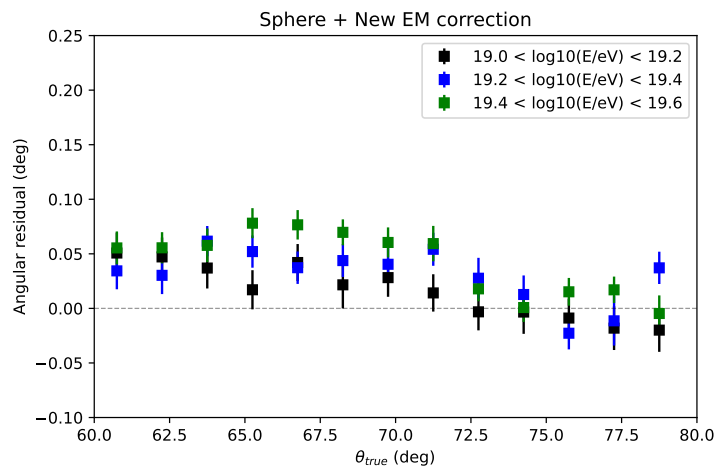
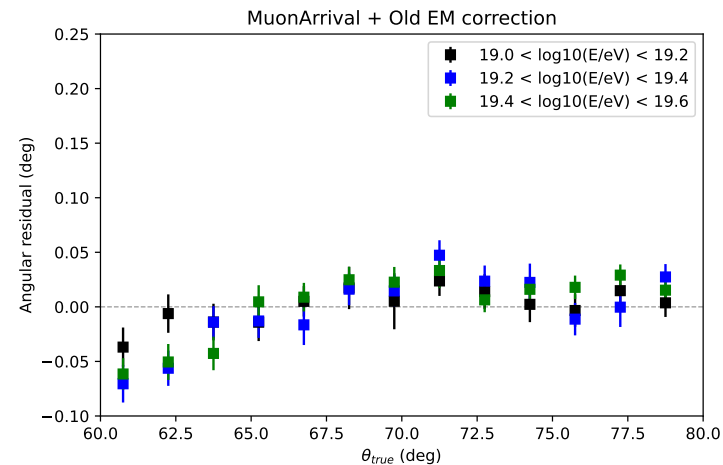
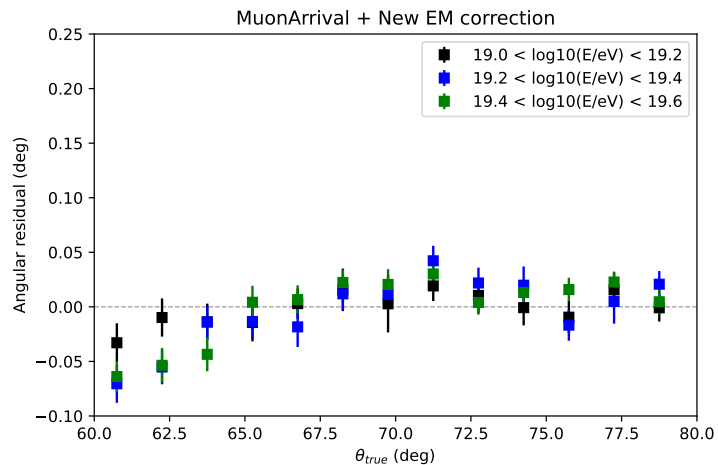
Peak at small p-values $\rightarrow \chi^2$ observed higher than expected,
model fails (bad reconstruction, insufficient cuts)



Biases changing models

- Two shower front models
(*Sphere and MuonArrival*)

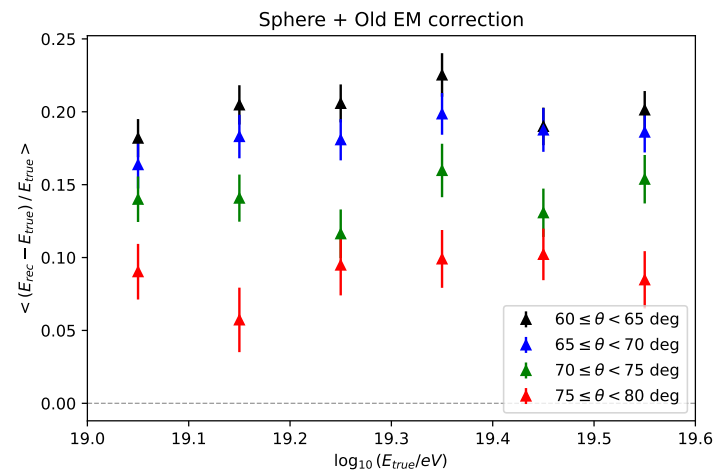
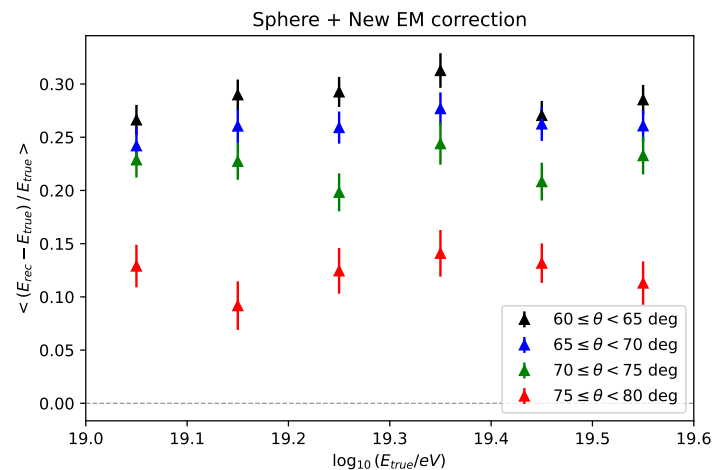
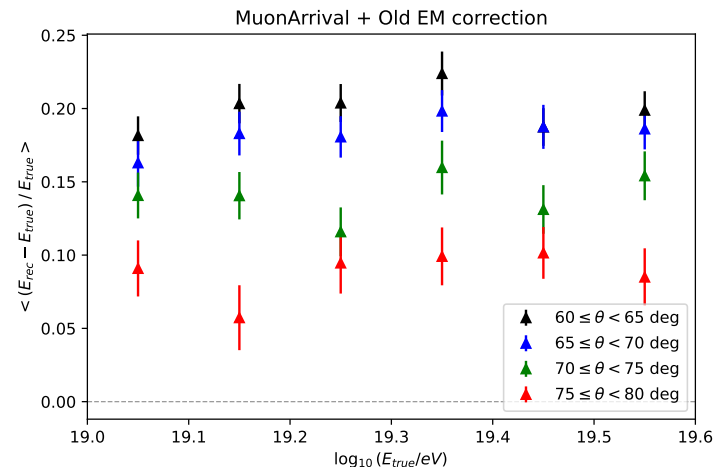
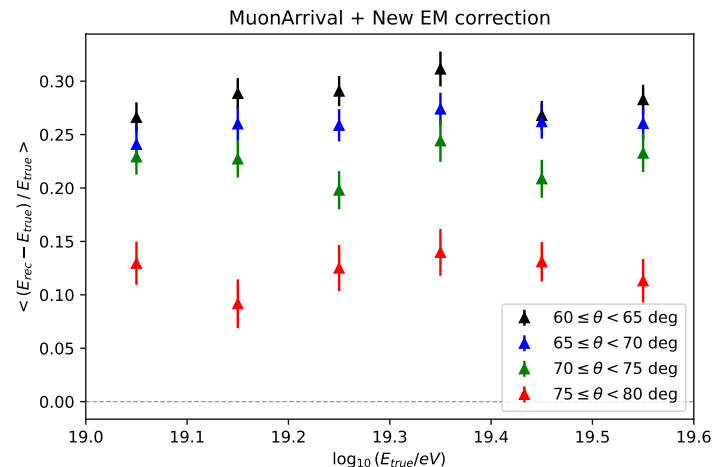
- Two EM corrections



Biases changing models

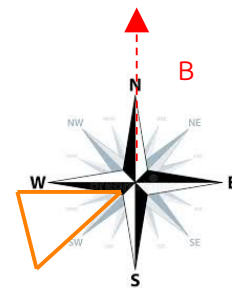
- Two shower front models
(*Sphere and MuonArrival*)

- Two EM corrections

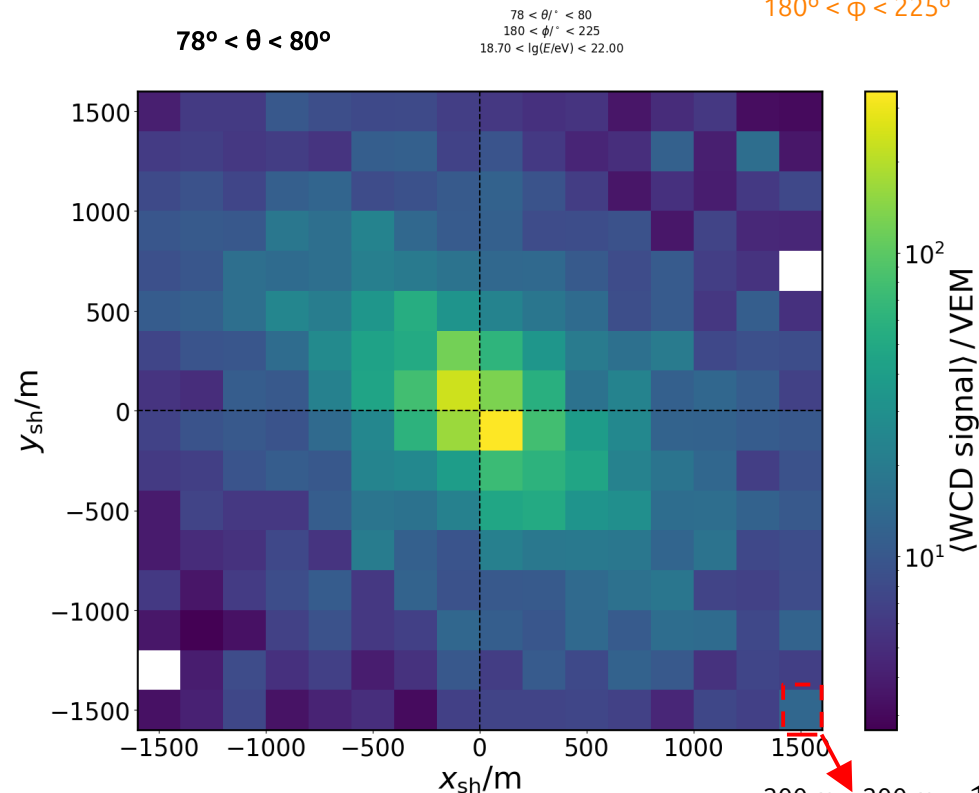
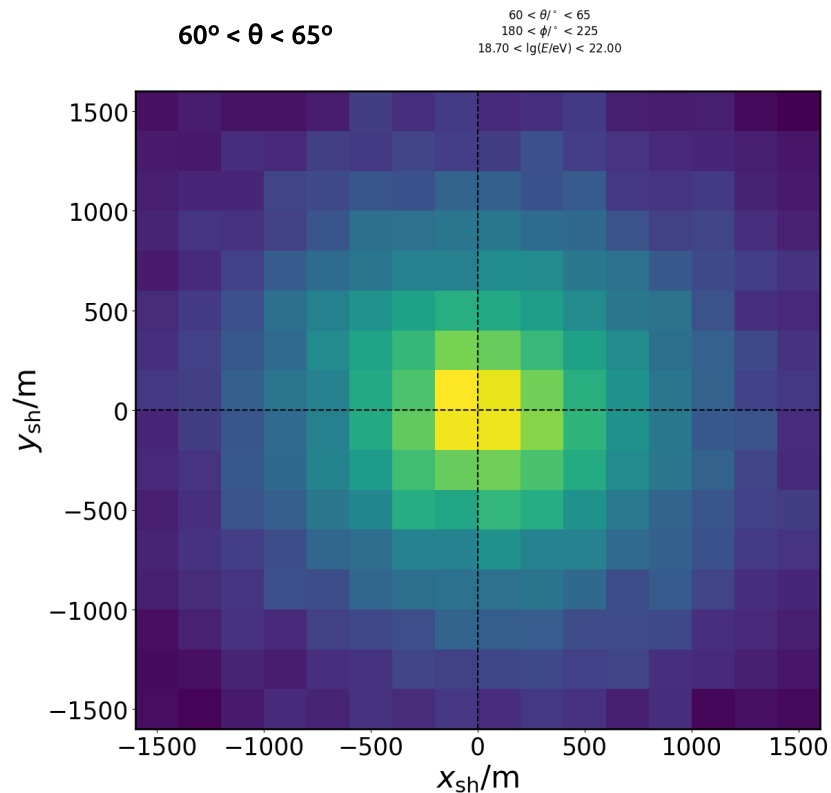


2D signal maps in HAS data

SD1500 Phase I → Average WCD signal in a “pixel” (x, y) in bins of (θ , ϕ) for $\lg(E/\text{eV}) > 18.7$

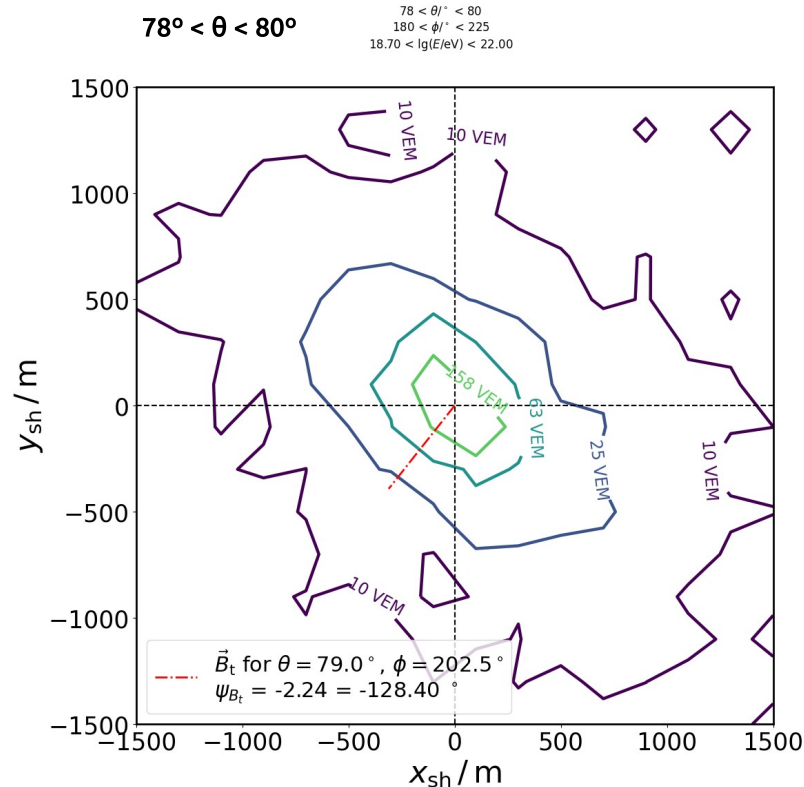
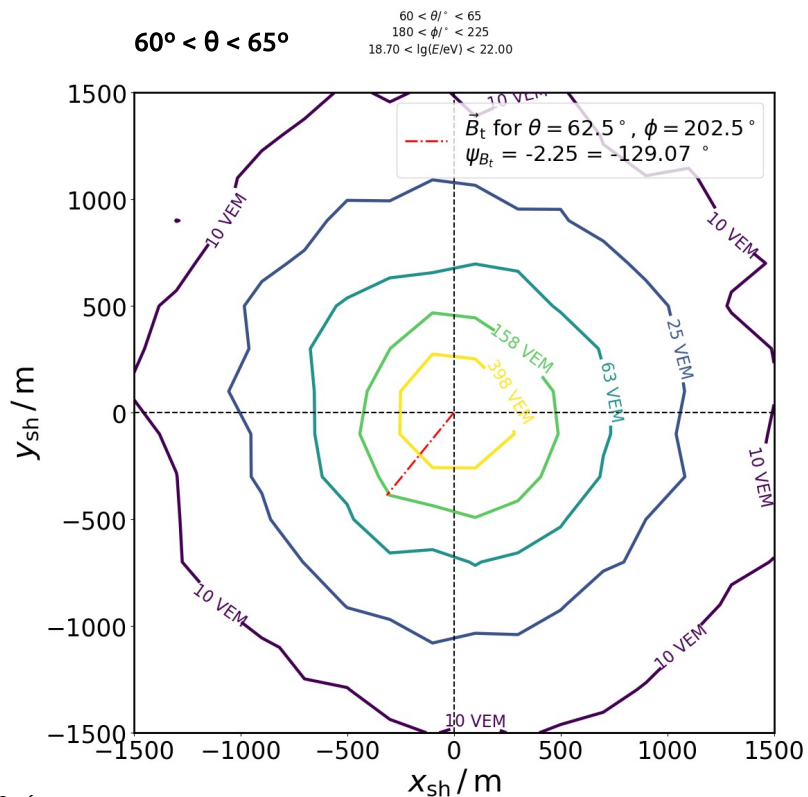
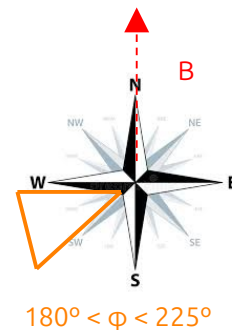


$180^\circ < \phi < 225^\circ$



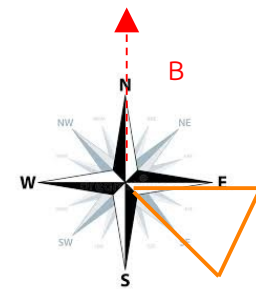
2D contours in HAS data

- Circular symmetry still visible for $60^\circ < \theta < 65^\circ$
- Symmetry lost for $78^\circ < \theta < 80^\circ \rightarrow$ Elongated in the direction perpendicular to B

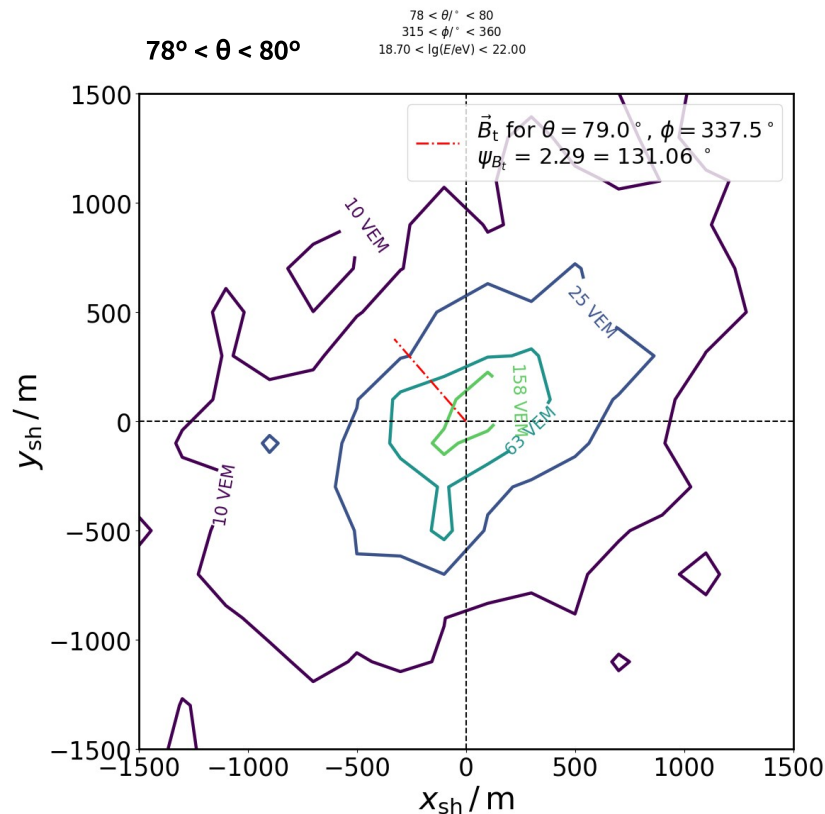
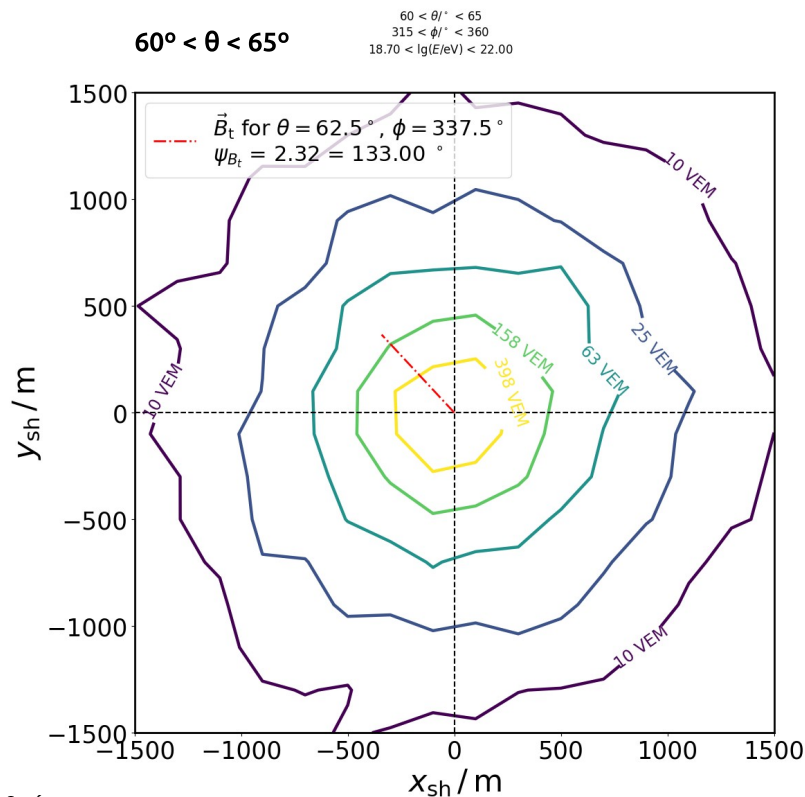


2D contours in HAS data

- Circular symmetry still visible for $60^\circ < \theta < 65^\circ$
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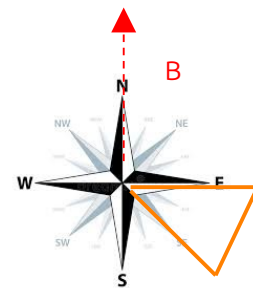


$315^\circ < \phi < 360^\circ$

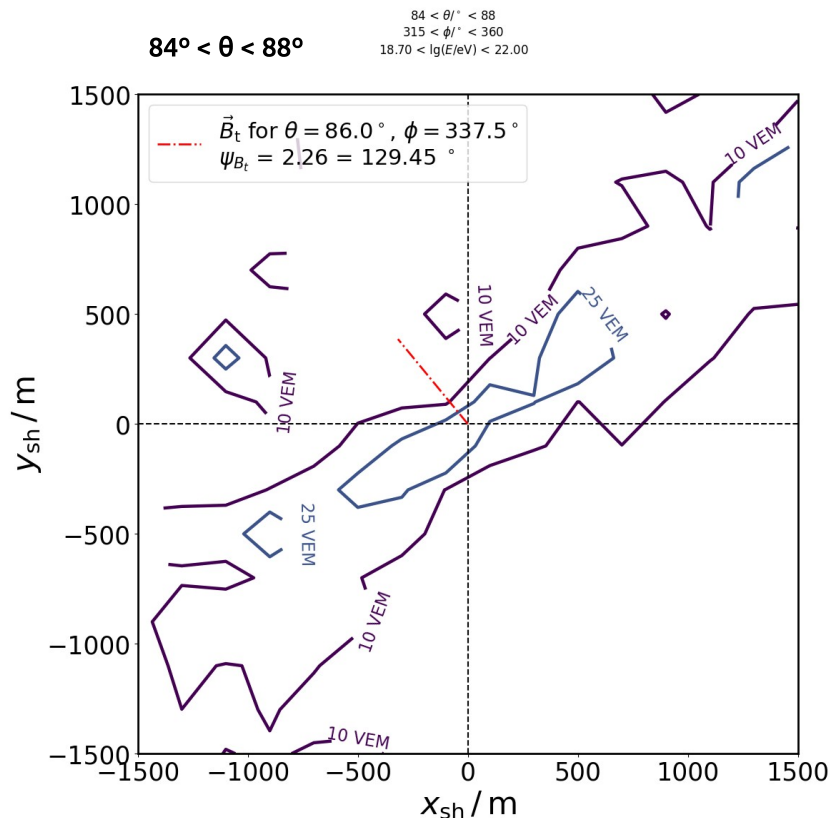
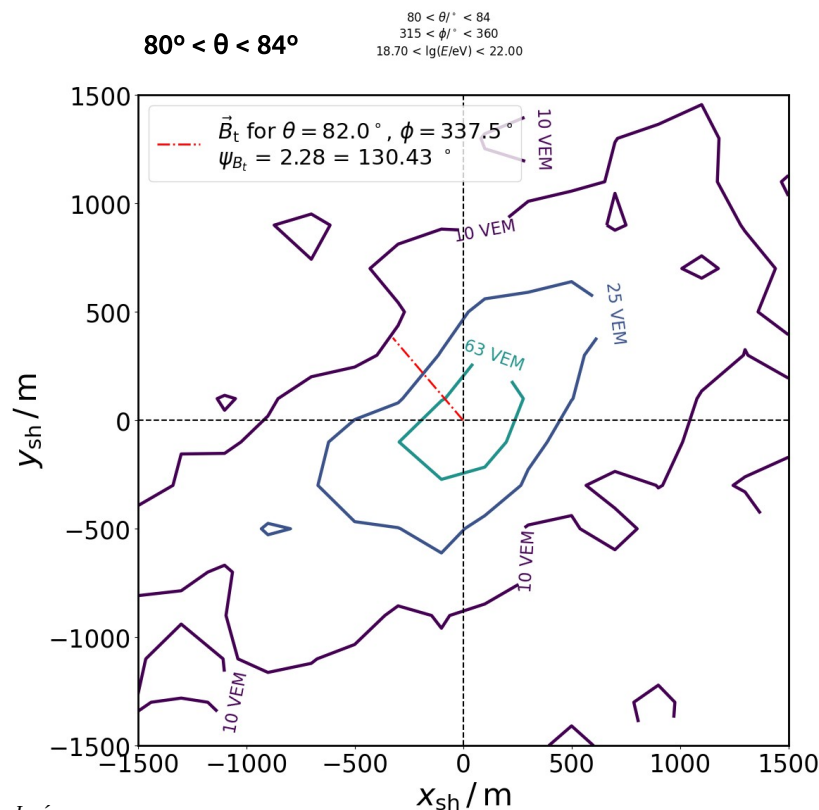


Extreme zenith angles

- Elongated pattern even more visible for $\theta > 80^\circ$
- Caveat: performance of reconstruction may not be optimal for these angles



$315^\circ < \phi < 360^\circ$



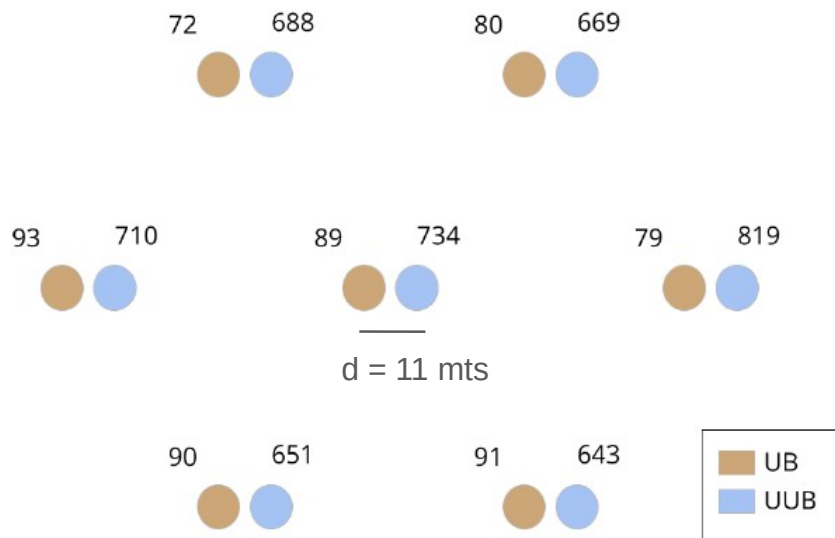
Validate Phase I and II HAS

Plan: carry out an event-by-event analysis similar to vertical showers using the UB/UUB doublet hexagon [GAP2023_032, GAP2023_033]

Low level:

- Distribution of mean and deviation of the VEM charge
- UB/UUB signal compatibility: For vertical events, signal S is corrected for radial dependence as: $S_{\text{corr}}(r) = S(r) - [S_{\text{LDF}}(r) - S_{\text{LDF}}((r1+r2)/2)]$
 - It is necessary to correct S for HAS?
 - θ dependence?
- AoP for phase I and phase II

Hexagon with doublet station UB-UUB

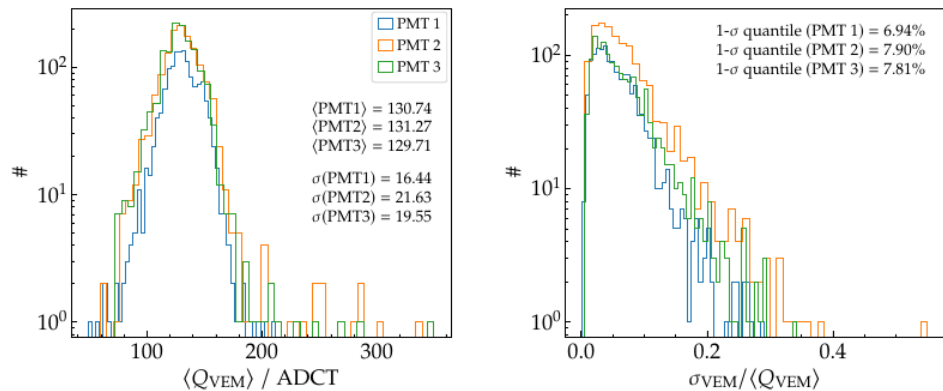


Event level:

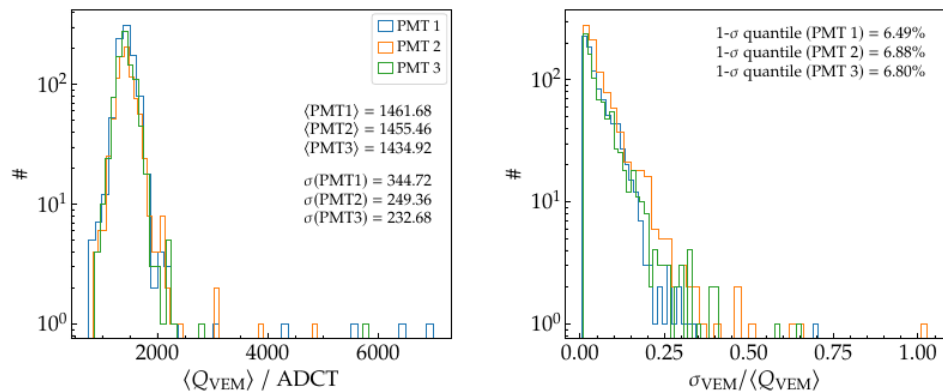
- # Rec events
- Core position
- N_{19} and energy calibration

Vertical showers [Allan Payeras]

UB WCD LPMTs

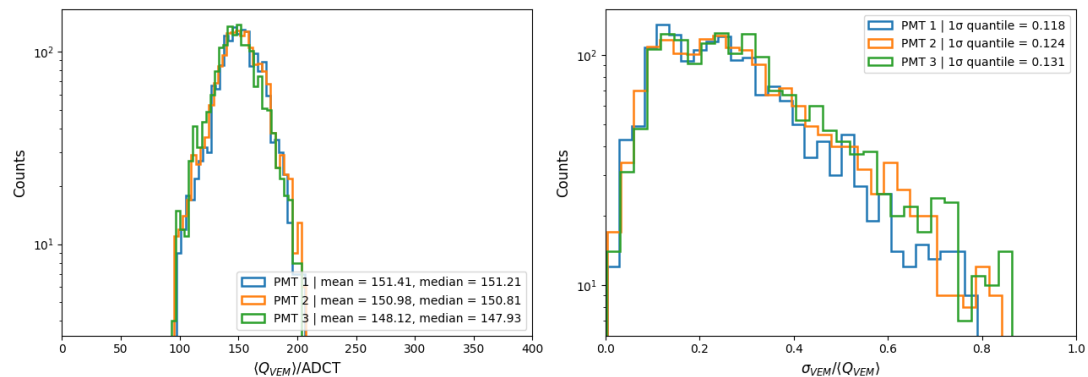


UUB WCD LPMTs

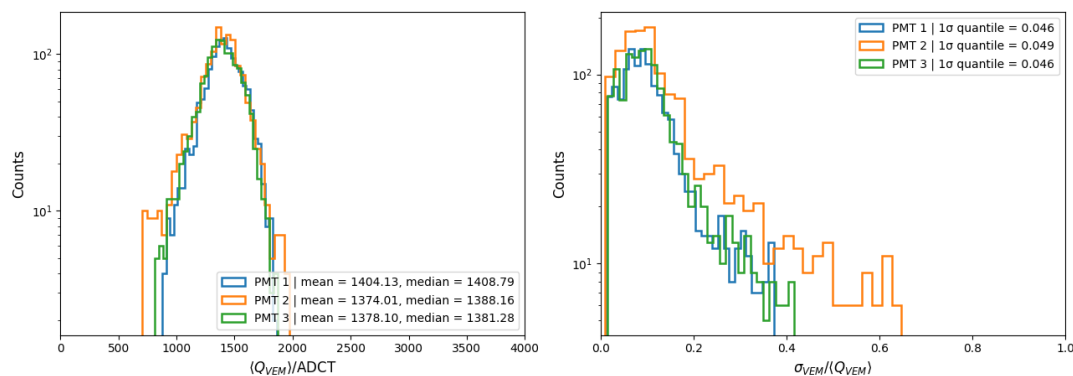


Inclined showers

UB WCD



UUB WCD



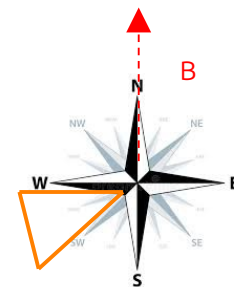
Next steps

- Finish HAS (cutting edge)
 - EM correction → expand beyond 70°
→ explore EM correction from SSD
 - CIC
 - SD750 and SD433
- Compute biases of N_{19} vs R_μ
- Compare muon maps data vs simulations (explore the potential to mass/hadronic models discrimination)
- Goodness of fit of current muon map for different primaries and hadronic models
- Study outliers
- Use UMD to revisit muon halo (contamination e^+ , e^-)

Backup Slides

2D contours – SD750

Same behavior observed for SD750, but a little bit more noisy (less statistics & less muons)



$180^\circ < \phi < 225^\circ$

